

HubForge OS

Open Source Decision Intelligence Infrastructure for Complex Systems

Build, Learn, Reason, Improve.

What is HubForge OS?

HubForge OS is an open-source operating system for building expert reasoning systems.

It enables organizations, researchers, institutions, developers, and communities to create intelligent systems capable of solving complex problems through recursive reasoning, structured knowledge retrieval, self-critique, continuous learning, and institutional memory.

Unlike conventional AI systems that generate outputs once, HubForge OS continuously improves outputs through iterative reasoning loops.

The goal is not simply generating answers.

The goal is building systems that think like experts.

Core Thesis

Most AI systems work like this:

Problem

→ Generate Response

→ Deliver Output

This produces shallow intelligence.

HubForge OS works differently.

Problem

→ Retrieve Knowledge

→ Generate Draft

→ Critique Logic

→ Improve Output

→ Evaluate Quality

→ Store Learning

→ Repeat Until Threshold Reached

This mirrors how real experts reason.

Why This Project Exists

Modern organizations increasingly face complex decision-making challenges.

Governments struggle with policy design.

Researchers struggle with synthesizing evidence.

Healthcare systems struggle with diagnostic reasoning.

Development organizations struggle with program evaluation.

Institutions repeatedly lose knowledge because expertise remains fragmented.

Existing software helps organizations collect and visualize information.

But software rarely helps organizations reason better.

HubForge OS exists to solve this problem.

Architecture Overview

HubForge OS consists of two layers.

Core Engine

- Reasoning Engine
- Retrieval Engine
- Rule Engine
- Critique Engine
- Evaluation Engine
- Memory Engine
- Loop Controller

Domain Packs

- Social Impact
- Public Policy
- Research Systems
- Healthcare
- Education
- Climate Intelligence

- Legal Systems
- Enterprise Decision Systems

The Core Engine remains universal.

Domain Packs encode specialized expertise.

Design Philosophy

Every output generated by HubForge OS follows one rule.

No output is trusted after first generation.

All outputs must pass through recursive refinement.

The system continuously asks:

- Is the reasoning correct?
- Is supporting evidence available?
- Are assumptions weak?
- Can the solution improve?
- Has similar reasoning been used previously?

Outputs improve through recursive loops.

Example Workflow

User Input:

Design a climate adaptation strategy for smallholder farmers.

System Process:

Understand Problem

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Retrieve Climate Frameworks

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Generate Strategy

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Critique Assumptions

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Improve Design

↓

Evaluate Feasibility

↓

Store Learning

↓

Repeat

Final output reflects expert-grade reasoning.

Principles

HubForge OS follows six principles.

Open Infrastructure

Decision intelligence should remain public infrastructure.

Evidence First

Systems must retrieve evidence before generating new reasoning.

Computational Efficiency

Expensive reasoning should be minimized whenever possible.

Transparency

Users should understand how reasoning decisions were made.

Continuous Learning

Systems must preserve institutional memory and improve over time.

Community Driven Development

Knowledge systems improve through collective contribution.

Current Development Stage

Early architecture design.

Currently building:

- Knowledge ontology
- Core reasoning architecture

- Agent runtime system
 - First prototype
 - Domain pack system
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Contributing

We are actively seeking contributors.

Needed expertise:

- AI engineering
- Knowledge systems design
- Research methodology
- Open source infrastructure
- Domain experts
- Decision science researchers

See CONTRIBUTING.md

Long Term Vision

HubForge OS aims to become universal infrastructure for expert reasoning systems.

A future where any organization can build systems that reason, critique, learn, and improve continuously.

The long-term goal is simple.

Make world-class decision intelligence universally accessible.

License

Apache 2.0

Build systems that think better.